
State of the Art Paper

The transient energy function method

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The status of the transient energy function (TEF) method of direct analysis of transient stability of power systems is reviewed. The mathematical model is presented; an elementary theoretical basis of the method is introduced; the procedure for using the method in assessing power system transient stability is outlined; and the present condition of the technique and the available computer program is given. The outstanding issues, current development work and new applications are then summarized.

Keywords: short-term system dynamics, transient stability, security assessment

I. Introduction

In recent years considerable progress has been achieved in power system transient stability assessment using direct methods based on energy functions. This progress has covered two broad fronts: (1) in development of the 'tool' used in direct transient stability assessment; and (2) in diversity of the type of applications analysed, and hence the associated problems addressed, by this tool.

While the transient energy function (TEF) used by the authors and their co-workers is based on that developed by Athay *et al.*^{1,2} in the late 1970s, significant improvements have been obtained in two main areas: (1) in identifying the system transient energy responsible for the separation of some generators from the rest of the system^{3,4}; and (2) in estimating the critical value of this transient energy for the particular disturbance under investigation⁵. We note that accurate determination of both is needed for correct stability assessment.

Investigation of industry-related applications by direct methods of transient stability analysis have revealed a host of issues which will have to be successfully dealt with if direct methods are to be seriously considered by the industry. Some of these relate to the type of disturbance investigated (e.g. simple faults, complex disturbance sequence, power limits, loss of generation, etc). The size of the power network used in the investigation can be important in revealing some dynamic problems which may not appear in smaller networks; in addition, analysis

of large networks may introduce computational difficulties⁶. The system loading level or the condition of the post-disturbance network may introduce other (mostly computational) difficulties⁷. Some applications require better power system models than that commonly used in direct stability analysis⁸. Associated with all these problems, the analyst must also deal with the questions of how accurate and how reliable the results obtained should be.

In this paper the status of the TEF method is reviewed; the mathematical model is presented; an elementary theoretical basis of the method is introduced; direct transient stability analysis using the method for different applications is outlined; and the present condition of the available software is given. The outstanding issues and current development work are then summarized.

Before proceeding with the review of the TEF method, we first offer some general remarks about power system transient stability. These remarks will help establish the context in which direct stability analysis is made.

II. Power system transient stability

Power system transient stability deals with the power system's ability, after it has been subjected to a large disturbance, to reach an acceptable steady-state condition⁹. A disturbance is a sudden change in a power system parameter or operating condition. A disturbance is considered large when linearization of the system equations, for the purpose of analysis, cannot be justified.

Transient stability studies are made by the electric utilities in North America to ascertain that criteria established by the North American Electric Reliability Council (NERC) are met¹⁰. These criteria state that when certain disturbances occur, no cascading outages in the interconnected power network must result. Different regions in North America establish their own disturbance criteria according to the conditions which prevail in their own high-voltage network.

To meet the NERC criteria for avoiding cascading outages, engineers in North American utilities conduct many transient stability studies: when new generation or transmission facilities are planned; and when these studies are warranted by new or changing operating

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conditions. The latter factor represents a rather profound change in the utilities' environment in North America, because of heavier transmission loadings and greater interchange of power among neighbouring networks, and because fewer new transmission facilities are being built. Transient stability studies are now being conducted routinely to ensure that under various operating conditions cascading outages will not result when certain credible contingent events occur. The type and scope of these transient stability studies are varied and their number is often large.

The above trend explains the increased interest in direct transient stability analysis by the industry, as well as the increased effort in investigating direct methods. It also offers important constraints for use of the method. Based on the authors' experience, if direct methods are to be widely adopted by the electric utility industry, they should: (1) deal with the actual industry problems, i.e. investigate the type of disturbances of interest to the industry; (2) provide the kind of answers required or needed by the industry (e.g. power limits versus critical clearing times); (3) offer some advantage over current (e.g. time solution) methods; and (4) ensure that the effort required to adopt them is justified by the gains to be achieved (e.g. if network equivalencing or a lot of new data are required). Even when all these conditions are satisfied, we must contend with the fact that 'old habits die hard'. An analyst accustomed to looking at swing curves may not easily accept that a set of numbers can give more information on the power system dynamic performance than the swing curves.

In the past, well-meaning researchers have made unjustified claims about what direct methods can offer the power system analyst. The authors wish to point out that their own experience indicates that the TEF method is a tool which promises to augment time solution in most applications, and replace it only in special applications such as online or near real time applications.

III. The mathematical model

III.1 Classical power system model

For the classical power system model (see Chapter 2 of Reference 11) the equations of motion of the synchronous generators, written with respect to an arbitrary synchronous frame, are given by

$$\begin{aligned} M_i \dot{\omega}_i &= P_i - P_{ei} \\ \dot{\delta}_i &= \omega_i \quad i = 1, 2, \dots, n \end{aligned} \quad (1)$$

where n is the number of generators,

$$P_{ei} = \sum_{\substack{j=1 \\ j \neq i}}^n [C_{ij} \sin(\delta_i - \delta_j) + D_{ij} \cos(\delta_i - \delta_j)]$$

$$P_i = P_{mi} - E_i^2 G_{ii}$$

$$C_{ij} = E_i E_j B_{ij} \quad D_{ij} = E_i E_j G_{ij}$$

and, for generator i ,

- P_{mi} mechanical power input
- G_{ii} driving point conductance
- E_i constant voltage behind the direct axis transient reactance
- ω_i, δ_i generator rotor speed and angle respectively

- M_i inertia constant
- $B_{ij}(G_{ij})$ transfer susceptance (conductance) in the reduced bus admittance matrix.

In the above, the power network has been reduced to the internal generator nodes. The generators are represented by constant voltage sources. The loads which are represented by constant impedances are included in the resulting admittance matrix after elimination of all other network nodes. The resulting network is shown schematically in Figure 1.

Transformation of equations (1) into the centre of inertia (COI) coordinates is done by defining the position of the centre of inertia by the equations

$$\begin{aligned} \delta_0 &= \frac{1}{M_0} \sum_{i=1}^n M_i \delta_i \\ \dot{\delta}_0 &= \frac{1}{M_0} \sum_{i=1}^n M_i \dot{\delta}_i \end{aligned} \quad (2)$$

where

$$M_0 = \sum_{i=1}^n M_i$$

The COI motion is defined by the equations

$$\begin{aligned} M_T \dot{\omega}_0 &= \sum_{i=1}^n P_i - P_{ei} \\ &= \sum_{i=1}^n P_i - 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^n D_{ij} \cos \delta_{ij} \triangleq P_{COI} \\ \dot{\delta}_0 &= \omega_0 \quad i = 1, \dots, n \end{aligned} \quad (3)$$

We now define the generators' angles and speeds relative to the COI by

$$\theta_i = \delta_i - \delta_0 \quad \tilde{\omega}_i = \dot{\delta}_i - \dot{\delta}_0 \quad i = 1, 2, \dots, n \quad (4)$$

The system equations of motion become

$$\begin{aligned} M_i \dot{\tilde{\omega}}_i &= P_i - P_{ei} - \frac{M_i}{M_T} P_{COI} \\ \dot{\theta}_i &= \tilde{\omega}_i \quad i = 1, \dots, n \end{aligned} \quad (5)$$

We note that the centre of inertia variables satisfy the constraints

$$\sum_{i=1}^n M_i \theta_i = \sum_{i=1}^n M_i \tilde{\omega}_i = 0 \quad (6)$$

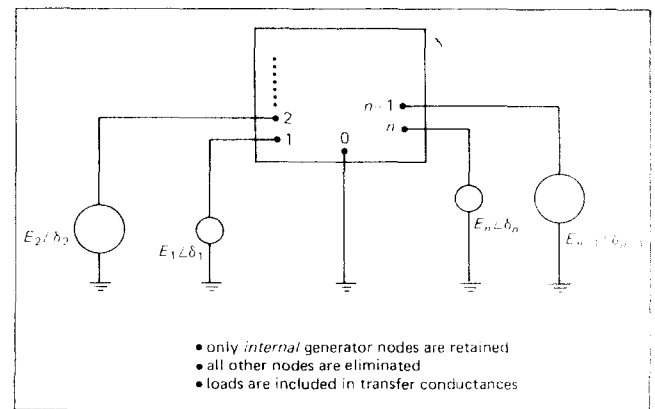


Figure 1. Classical model

III.2 The transient energy function

The transient energy function V is always defined for the post-disturbance system. It can be derived from the n acceleration equations in the COI frame of reference (equations (5)). At a given system state it is given by

$$V = \sum_{i=1}^n \frac{1}{2} M_i \tilde{\omega}_i^2 - \sum_{i=1}^n P_i (\theta_i - \theta_i^s) - \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left(C_{ij} (\cos \theta_{ij} - \cos \theta_{ij}^s) + \int_{\theta_i^s + \theta_j^s}^{\theta_i + \theta_j} D_{ij} \cos \theta_{ij} d(\theta_i + \theta_j) \right) \quad (7)$$

We note that the terms of the transient energy are evaluated relative to the post-disturbance stable equilibrium conditions ($\theta = \theta^s, \tilde{\omega} = 0$). These terms are physically interpreted as follows¹:

- $\frac{1}{2} \sum_{i=1}^n M_i \tilde{\omega}_i^2 = \frac{1}{2} \sum_{i=1}^n M_i \omega_i^2 - \frac{1}{2} M_T \omega_0^2$
total change in rotor kinetic energy relative to COI;
- $\sum_{i=1}^n P_i (\theta_i - \theta_i^s) = \sum_{i=1}^n P_i (\delta_i - \delta_i^s) - \sum_{i=1}^n P_i (\delta_0 - \delta_i^s)$
change in rotor position energy relative to COI;
- $C_{ij} (\cos \theta_{ij} - \cos \theta_{ij}^s)$
change in magnetic stored energy of branch ij ;
- $\int_{\theta_i^s + \theta_j^s}^{\theta_i + \theta_j} D_{ij} \cos \theta_{ij} d(\theta_i + \theta_j)$
change in dissipated energy of branch ij .

These terms represent energy dissipated in the transfer conductances and are naturally in the form of an integral which is path-dependent.

III.3 Modification of the transient energy function

Two modifications are made in the transient energy function expression given in (7).

- An approximation of the last term is used to avoid calculating the actual system trajectory. A linear trajectory in the angle space is assumed to allow the transfer conductance terms to be analytically evaluated between the limits θ^s and θ :

$$\sum_{i=1}^{n-1} \sum_{j=i+1}^n \int_{\theta_i^s + \theta_j^s}^{\theta_i + \theta_j} D_{ij} \cos \theta_{ij} d(\theta_i + \theta_j) \triangleq \sum_{i=1}^{n-1} I_{ij} \quad (8)$$

where

$$I_{ij} = D_{ij} \frac{\theta_i + \theta_j - \theta_i^s - \theta_j^s}{\theta_{ij} - \theta_{ij}^s} (\sin \theta_{ij} - \sin \theta_{ij}^s) \quad (9)$$

is an approximated term to account for the transfer conductances.

- It was found^{3,4} that not all the transient kinetic energy contributes to the separation of the critical (or severely disturbed) generators from the rest of the system. The corrected kinetic energy is that of two

equivalent groups of generators: the so-called critical group and the rest of the generators. This is computed as follows.

Let $\tilde{\omega}_{cr}$ and $\tilde{\omega}_{sys}$ be the speed of the inertial centres of the critical group and of the rest of the generators respectively; and let M_{cr} and M_{sys} be the inertia constants of the critical group and of the rest of the generators respectively. Then

$$M_{eq} = M_{cr} \cdot M_{sys} / (M_{cr} + M_{sys})$$

$$\tilde{\omega}_{eq} = \tilde{\omega}_{cr} - \tilde{\omega}_{sys}$$

and the corrected kinetic energy is given as

$$V_{KE|corr} = \frac{1}{2} M_{eq} \tilde{\omega}_{eq}^2 \quad (10)$$

IV. Stability analysis

IV.1 Asymptotic stability

For stability analysis, a power system is an autonomous system described by

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$$

The equilibrium point $\mathbf{f}(\mathbf{0}) = \mathbf{0}$ is located at the origin. The origin is said to be *asymptotically stable* if

- stable,
- $\lim_{t \rightarrow \infty} \|\mathbf{x}(t)\| \rightarrow 0$ for motion starting sufficiently close to the origin.

Note. In power system transient stability, asymptotic stability is of interest.

IV.2 Lyapunov method

The basic idea is summarized as follows:

- if $E(\mathbf{x})$ is the system energy, and if dE/dt is negative except at equilibrium, then $E(\mathbf{x})$ will continue to decrease until it assumes the value $E(\mathbf{x}_e)$, or $E(\mathbf{0})$ when the equilibrium is at the origin;
- in the mathematical treatment $E(\mathbf{x})$ is replaced with the scalar function $V(\mathbf{x})$ which has certain properties.

The above ideas are incorporated in the following: if there exists $V(\mathbf{x})$ such that

$$V(\mathbf{x}) > 0 \quad (\text{positive definite})$$

$$\dot{V}(\mathbf{x}) < 0 \quad (\text{negative definite})$$

then the origin is *asymptotically stable*.

IV.3 Region of asymptotic stability

Consider the autonomous system

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) \quad \mathbf{f}(\mathbf{0}) = \mathbf{0}$$

Let $V(\mathbf{x})$ be a scalar function and let Ω be a region where $V(\mathbf{x}) > 0$ and $\dot{V}(\mathbf{x}) < 0$ in Ω ; then all motions starting in Ω to converge to the origin at $t \rightarrow \infty$.

We note that successful application of the Lyapunov

method to power systems requires, in addition to the construction of a suitable V -function, making a good estimate of the region of stability Ω . Much of the conservativeness of the early work on application of the Lyapunov method to power systems was due to estimation of a region of stability which is too small for correct stability assessment.

A great deal of the progress in this area was associated with the development of the TEF method. By using heuristic arguments, physical reasoning and plain engineering sense, more realistic estimates were made of the region of stability and hence of the critical energy against which transient stability assessment can be made¹⁻³.

Theoretical progress, which caught up with the above, was made via invoking the invariance principle, first introduced by LaSalle (see e.g. Reference 12). Reference 1 first used the invariance principle to illustrate the concept of the controlling UEP (see Section IV.5 below). The authors of Reference 13 also used the invariance principle to validate the concept of the individual machine energy function.

By invoking the invariance principle, an important restriction was removed by requiring that the derivative of V need only be semidefinite, i.e.

$$\dot{V}(\mathbf{x}) \leq 0$$

IV.4 Present theory on characterization of the stability boundary¹⁴

Let $\mathbf{x}_s$ be the stable equilibrium point, $\mathbf{x}_i$ be the unstable equilibrium points ($i = 1, 2, \dots$), $\partial A(\mathbf{x}_s)$ be the stability boundary of $\mathbf{x}_s$ and $W^s(\mathbf{x}_i)$ be the stability manifold of the unstable equilibrium point $\mathbf{x}_i$; then

$$\partial A(\mathbf{x}_s) = UW^s(\mathbf{x}_i) \quad \mathbf{x}_i \in E \cap \partial A$$

Let $\mathbf{x}_e$ be the point where the unstable trajectory $\mathbf{x}(t)$ crosses the stability boundary $\partial A(\mathbf{x}_s)$, i.e. $\mathbf{x}_e$ is the exit point of trajectory $\mathbf{x}(t)$. If $\mathbf{x}_e$ is 'close' to $\mathbf{x}_i$, then $V(\mathbf{x}_e)$ is the true critical value of $V(\mathbf{x})$.

Let $\mathbf{x}_{co}$ be a UEP on the stability boundary and let $\mathbf{x}_e$ belong to the stable manifold $W^s(\mathbf{x}_{co})$ of the UWP $\mathbf{x}_{co}$. Since $V(\mathbf{x})$ assumes its minimum on the stable manifold $W^s(\mathbf{x}_{co})$ at $\mathbf{x}_{co}$, then the unstable trajectory must pass through the constant energy surface

$$\{\mathbf{x}: V(\mathbf{x}) = V(\mathbf{x}_{co})\}$$

For a given unstable trajectory $\mathbf{x}(t)$, if the exit point $\mathbf{x}_e$ lies on the stable manifold of $\mathbf{x}_{co}$, then we call $\mathbf{x}_{co}$ the controlling UEP (for this disturbance).

IV.5 Example: a three-generator system¹

Reference 1 gives a detailed discussion of the region of stability of a three-generator system, shown in Figure 2. The potential energy surfaces are given in Figure 3, which shows the SEP, the different UEPs and the potential energy boundary surface (PEBS) for that system. The concept of the controlling UEP (for a given disturbance) and its significance are illustrated in Figure 4.

In Figure 4 the faulted trajectory is depicted by the solid trajectory emanating from the SEP and post-fault trajectories are shown by the dashed lines, each

corresponding to different clearing times. The critical clearing time, i.e. the fault clearing time t_c^* such that for all fault clearing times t less than t_c^* the system remains transiently stable, is depicted on the fault trajectory in Figure 4. The post-fault trajectories $S_1, \dots, S_n, \dots$ are stable, while $\dots, u_1, u_2, \dots$ are unstable.

Let us now describe how the theorem presented previously can be conceptually applied to the problem in order to determine a critical clearing time. We shall define the invariant set Ω for the post-fault system in the following way. First, note that the SEP is an invariant set. So is any one of the post-fault trajectories starting from initial conditions which fall on the faulted trajectory. Now, the union of all such post-fault trajectories with the SEP forms the invariant set of interest, Ω . Let the real-valued function $V(\mathbf{x})$ correspond to the transient energy of the post-fault system. Then it is not difficult to see that the value of the largest fault-on gained energy along the fault trajectory is that which allows for both the conditions on $V(\mathbf{x})$ and $\dot{V}(\mathbf{x})$ of the stability theorem to be satisfied. In Figure 4 it is assumed that this value yields the critical clearing time value of $\bar{t}_c$. Note that in general $\bar{t}_c$ is expected to be less than the true critical clearing time t_c^* because the theorem provides sufficient but not necessary conditions for stability.

The case depicted in Figure 4 corresponds to a first-swing instability where the critically unstable trajectory escapes almost through (or very close by) the appropriate critical UEP. The latter behaviour for first-swing cases is based on and consistent with observations made for first-swing instabilities from numerous and exhaustive multimachine case studies. The trajectories $S_1, \dots, S_n$, etc are stable and hence do not cross the PEBS, in accordance with the proposed instability conjecture, while the trajectories $u_1, \dots, u_n$, etc are unstable and, along with the fault-on trajectory itself, cross the boundary surface from the inside.

V. Procedure for transient stability assessment using the TEF method

V.1 Transient stability

Conceptually, the procedure for transient stability assessment using the TEF method is rather simple. Two values of the transient energy V are computed and compared:

- the value of V when the disturbance is terminated,

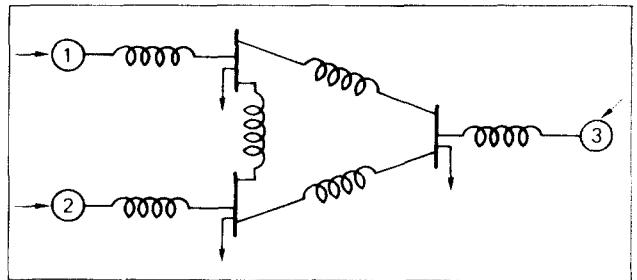


Figure 2. Three-machine system example

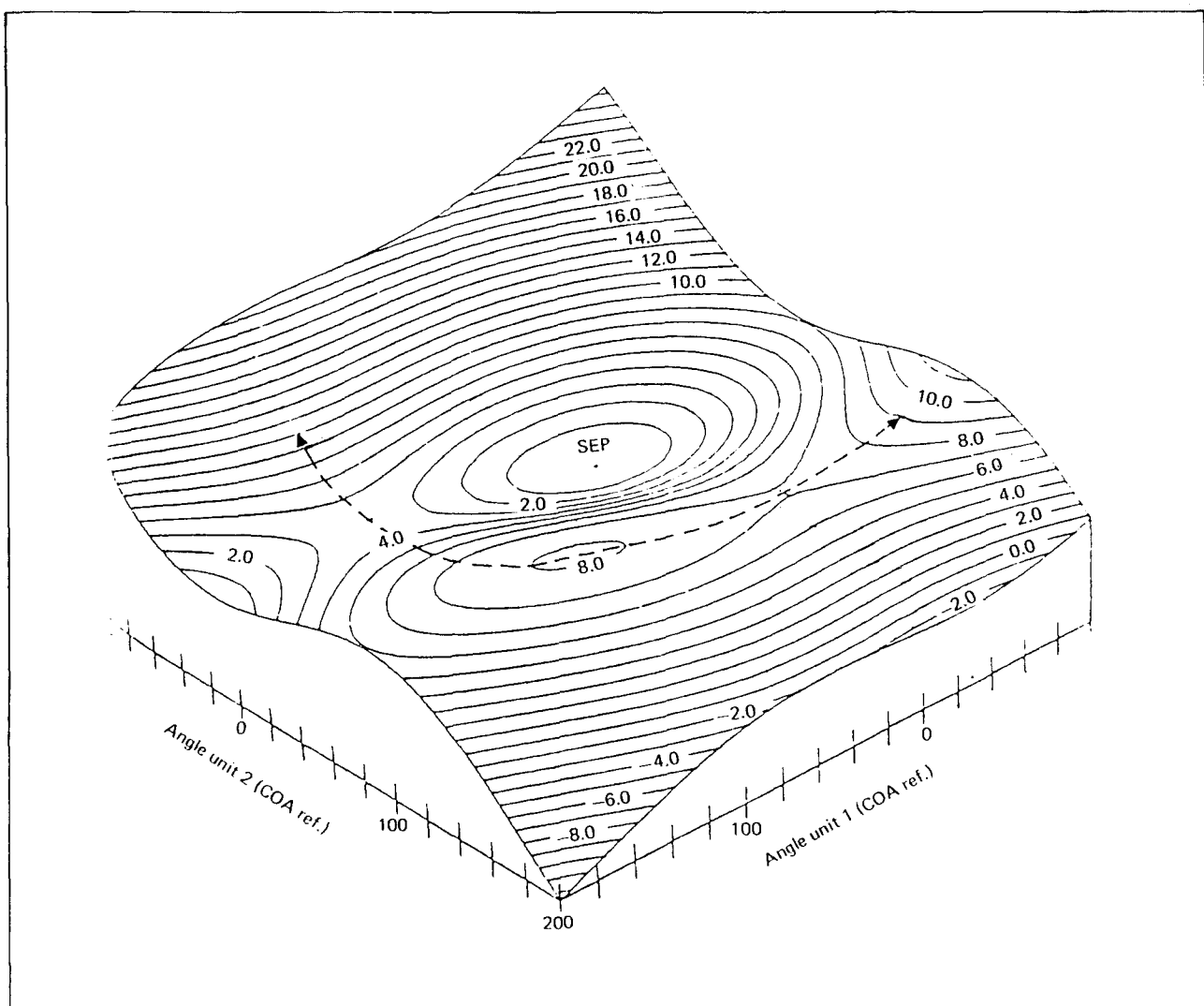


Figure 3. Potential energy function for three-machine system

- e.g. if the disturbance is a simple fault, the value of V at fault clearing, V_{cl} ;
- the critical value of V , V_{cr} , which is the potential energy at the controlling UEP for the particular disturbance under investigation.

Transient stability assessment is made by comparing these two values: if $V_{cl} < V_{cr}$, the system is stable; and if $V_{cl} > V_{cr}$, the system is unstable.

Alternatively the assessment is made by computing the energy margin ΔV given by

$$\Delta V = V_{cr} - V_{cl} \quad (11)$$

Transient stability (or instability) is decided according to whether $\Delta V \geq 0$.

Transient stability assessment using the energy margin has two advantages: (i) the energy margin can be obtained by one computation only, by evaluating V between the conditions at fault clearing (θ^{cl}, ω^{cl}) and those at the UEP ($\theta^u, 0$); and (ii) a qualitative measure of the degree of stability (or instability) can be obtained if ΔV is normalized³ with respect to the corrected kinetic energy at fault clearing, i.e.

$$\Delta V|_{norm} = \Delta V / V_{KE|corr} \quad (12)$$

The above procedure is simple enough. However, complications arise when the procedure is applied to practical power networks. These complications will be dealt with in later sections. In the meantime, the steps required in the TEF procedure will be outlined.

V.2 Main steps in the TEF procedure

V.2.1 Initial computations These calculations involve reading of the pre-disturbance load flow data and the appropriate generator dynamic model data.

If the classical power system model is to be used, the generator parameters needed are the inertia constant H and the direct axis transient reactance x'_d for each generator. The following computations are then made for the various generators: the voltage behind x'_d , the mechanical power and the initial generator angle θ^{s1} in the inertial centre reference frame. From the initial power, power factor and voltage at each load bus, the equivalent constant impedances for the various loads are computed.

If for some generators higher-order models are used, the appropriate generator and exciter parameters are needed: for the generators, x'_d , x'_q , τ'_{d0} , τ'_{q0} and H ; for the exciter, a one-gain, one-time-constant, one-limiter model is used. The initial values of E'_d and E'_q are also needed.

network is steady-state unstable, or no solution exists. In this case direct stability analysis cannot be made.

V.2.4 Identifying the controlling UEP The UEP is a steady-state system condition which is never reached by the system. It merely separates the stable and unstable system trajectories (as shown in Figure 4). It is a solution of the steady-state system equation (13) with certain generators' angles generally greater than $\pi/2$; this is why the solution is unstable. The potential energy at the UEP represents the power network's ability to 'absorb' all the transient energy at the end of the disturbance, i.e. to convert all the kinetic energy into potential energy.

Different UEPs have different generators' angles generally advanced ($> \pi/2$). Thus different UEPs have different potential energies. It is important therefore to correctly identify the controlling UEP for the disturbance under investigation, and compute the value of its potential energy. This is the critical value of V against which transient stability assessment is made, and is one of the key steps in the TEF procedure.

Identification of the controlling UEP is made by the so-called 'mode of disturbance' (MOD) test⁵. In that test the UEP with the lowest normalized potential energy margin (ΔV_{PEn}) at the instant the disturbance is removed is identified. This test is applied to a set of candidate UEPs in the direction of the (disturbed) system trajectory:

$$\Delta V_{PEn} = (V_{PE}^u - V_{PE}^{cl}) / V_{KE}|_{corr} \quad (14)$$

To simplify the computations involved in the MOD test, ΔV_{PEn} is computed, for the candidate UEP modes, at a point *near* the UEP. When the UEP of interest is identified, the actual UEP is computed.

V.2.5 Determination of the UEP and V_{cr}

- For the given mode, i.e. with certain generators' (critical) angles advanced, compute the corner point $\hat{\theta}^u$. For the critical generators $\hat{\theta}_i^u = \pi - \theta_i^s$; and for the rest of the generators $\hat{\theta}_i^u = \theta_i^s$.
- The values of $\hat{\theta}^u$ are adjusted to account for the motion of the inertial centre from θ^s to $\hat{\theta}^u$, to make sure that the condition $\sum_{i=1}^n M_i \theta_i = 0$ is satisfied (see Section 3.3.2 of Reference 16).
- The point on the ray from θ^s to $\hat{\theta}^u$ at which the potential energy is maximum is found. This is called θ^{ray} . θ^{ray} is used in the MOD test, and when the candidate UEP with the lowest ΔV_{PEn} is identified, it is also used as the starting point for the UEP computation.
- Starting from θ^{ray} , the controlling UEP (θ^u) is solved for (13): (i) first, a modified NR method with cubic interpolation in the iterative procedure and scaling technique is used; and (ii) if the NR method does not converge, a robust corrected Gauss-Newton method is used instead^{7,16}.
- The potential energy corresponding to θ^u is then computed. This is the V_{cr} used in transient stability assessment.

The TEF procedure described above is summarized in the flowchart shown in Figure 5.

VI. Status of TEF method used with classical power system model

In the EPRI Project RP2206-1, transient stability assessment using the TEF method was extensively tested on several power systems derived from parts of the Ontario Hydro power network. The systems varied in size from 11 generators to over 200 generators, and from 50 buses to over 1600 buses. They also differed in the types of stability problems investigated (e.g. stability classification, power limits, etc), and in the dynamic system behaviour (e.g. plant mode, inter-area mode, etc). In addition to these series of tests, the authors and their co-workers have been involved in investigating the use of the TEF method with three other networks, derived from actual power networks in North America, which varied in size from 17 generators to 25 generators, and from 50 buses to over 200 buses. The classical power system model has been used in all these tests, which involved over 2500 transient stability runs. In these tests the transient stability results obtained by the TEF method were compared with similar results, for the same system model and conditions, obtained by time simulation. The following summary is based on these results.

VI.1 Stability classification

Excellent first-swing stability analysis capability has been demonstrated throughout. This includes the following information.

- Very good prediction of stability classification. This means that if critical clearing time is of interest, TEF gives a good estimate of it. If stability assessment is made at other fault disturbances, the value of the normalized energy margin gives an indication of the degree of stability or instability.
- Other valuable information about the dynamic system behaviour is given by TEF analysis. In addition to the degree of stability, the TEF procedure gives the advanced generators (in the MOD test). These generators are the ones most severely affected by the disturbance. This information is helpful to the analyst for corrective action planning.
- Good stability classification has been achieved for both *plant* mode stability and *inter-area* mode stability results. To illustrate the difference, a sample result, taken from EPRI Report EL-4980¹⁶, is given. A single-line diagram of the Ontario Hydro Nanticoke study area is shown in Figure 6. Faults near the Nanticoke plants are investigated for two different loading conditions at the *Bruce* plants near by: (i) low Bruce generation (about 1600 MW); and (ii) high Bruce generation (over 3600 MW). In the former case, called 'Nanticoke normal system', when instability occurs, only the few plants near the fault are affected, as shown in Figure 7. In the latter case, when instability occurs, a system split involving 28 generators occurs. To illustrate this case very clearly, the rotor angles of interest are plotted in the COI reference frame in Figures 8(a) and 8(b). The angles of the Nanticoke generators (θ_{20} , θ_{26} and $\theta_{20,26}$) are shown together with the inertial centre of the 28 generators losing synchronism with the rest of the system when instability occurs. The inter-area mode is clearly demonstrated in these plots.

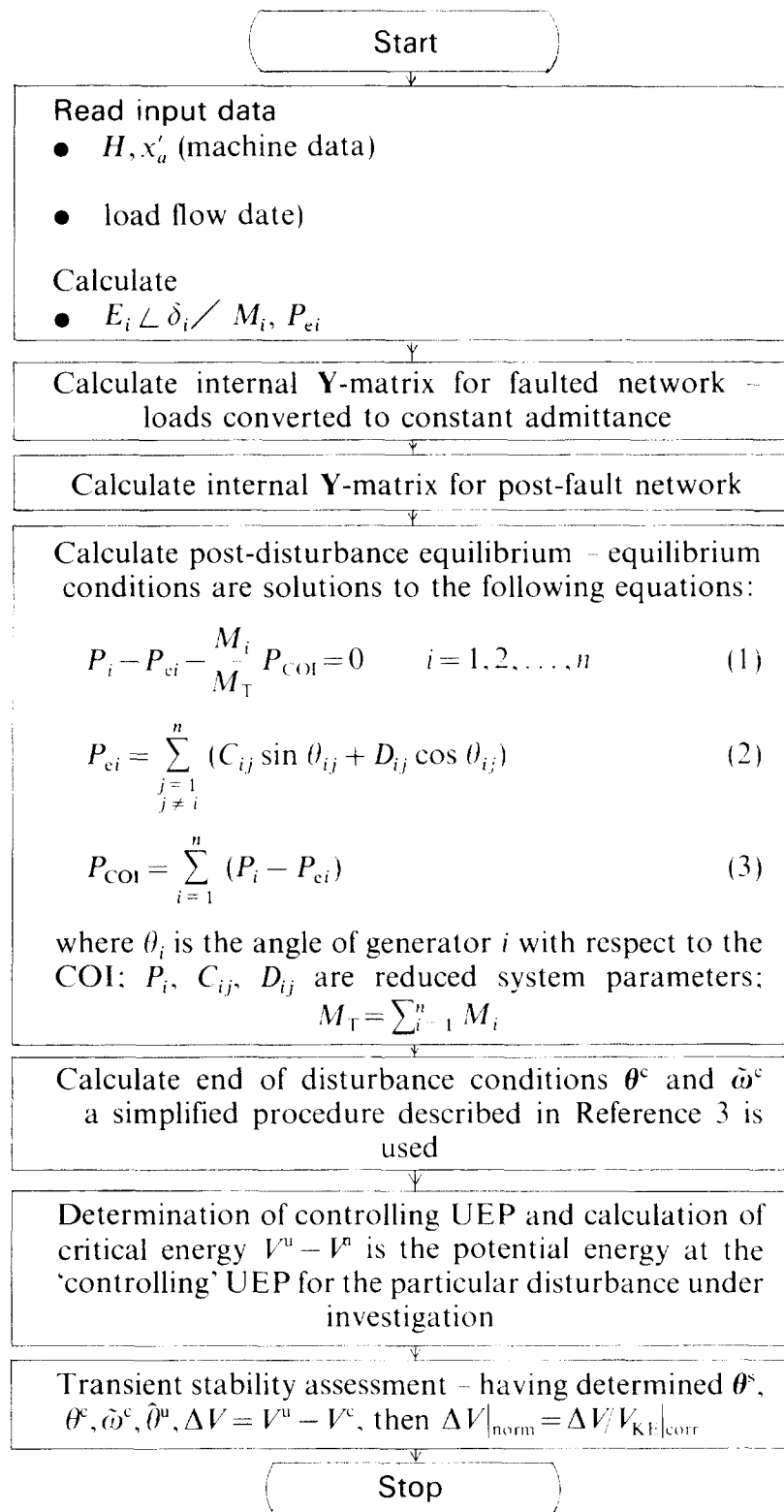


Figure 5. Flowchart for stability assessment calculations

It is to be emphasized that in the few cases in which the inter-area mode was encountered, the TEF procedure correctly predicted the resulting system split.

VI.2 Stability-limited power

Some cases were extensively investigated by the TEF method in the RP2206-1 project. In evaluating the method, the report concludes that '... the power limits determined by the TEF solution agreed with those

obtained by the time solution, within an accuracy of better than 5% in the majority of cases'.

The results of the studies conducted on five OH networks are summarized in Table 1. With the exception of the 50-generator Bruce system, the error in the predicted power limit is under 5%. The report¹⁶ attributes the 7.6% error to a convergence problem in the solution technique, and to the use of a large time step in the faulted network solutions. These errors were

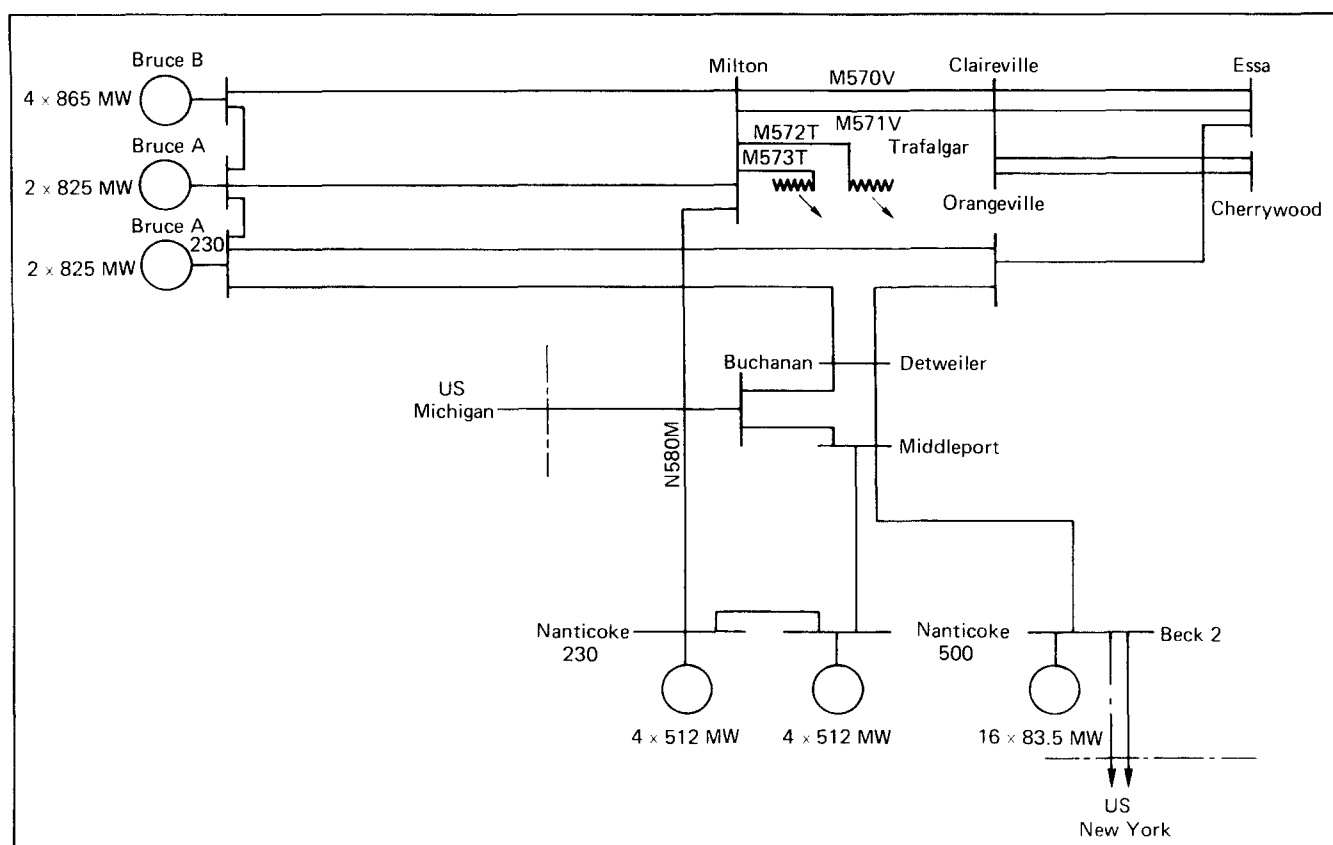


Figure 6. Single-line diagram of the Nanticoke area

substantially reduced in later versions of the TEF program.

VI.3 Solution speed

It should be emphasized that comparing computer solution times for conventional transient stability programs with that obtained with TEF is rather risky, since the results may not be equivalent for both. Furthermore, the TEF CPU time will not only depend on the type of computer used, but will also depend on the system condition (e.g. if the SNR or CGN method of solution is used). For these reasons, the data presented in Table 2 should be treated as examples of what is expected under 'normal' conditions. They are quoted in the TEF 1.1 Version User's Manual released in 1987 and are for a VAX 8600 computer.

VI.4 Variety of applications

The data presented above show that the TEF method has been extensively used for transient stability analysis of simple fault disturbances to obtain critical clearing times or power limits. In addition, the following TEF applications have been successfully investigated by researchers at Iowa State University, and by colleagues at Ontario Hydro, BC Hydro and Florida Power and Light Co:

- use of a linearized sensitivity technique to obtain power flow limits at a radial transmission interface¹⁷;
- use of linearized sensitivity analysis method to obtain stability-limited plant loading¹⁸;
- investigation of loss of generation transients, and

correlation of the transient energy margin to the apparent impedance of out-of-step relays^{15,19};

- stability analysis of a complex disturbance sequence to determine the amount of generation to be tripped to maintain stability²⁰;
- transmission expansion plans in system planning applications to investigate the effect of different transmission reinforcement plans¹⁶.

VI.5 The TEF computer program

The current publicly available version of the TEF computer program (early 1988) was released by the EPRI Software Center in August 1987. It is being tested by some users. A summary of its features is outlined below (see also Table 2).

VI.5.1 Main features and size

- classical model;
- for offline use;
- available in VAX, IBM and PRIME versions;
- dimensioned for 150-generator, 1500-bus power network.

VI.5.2 TEF program functions To implement the transient stability assessment, the current TEF program performs the following tasks for a given fault disturbance:

- obtains the power network internal Y-matrix (i.e. reduced to the internal generator nodes) for the faulted network and the post-fault network;

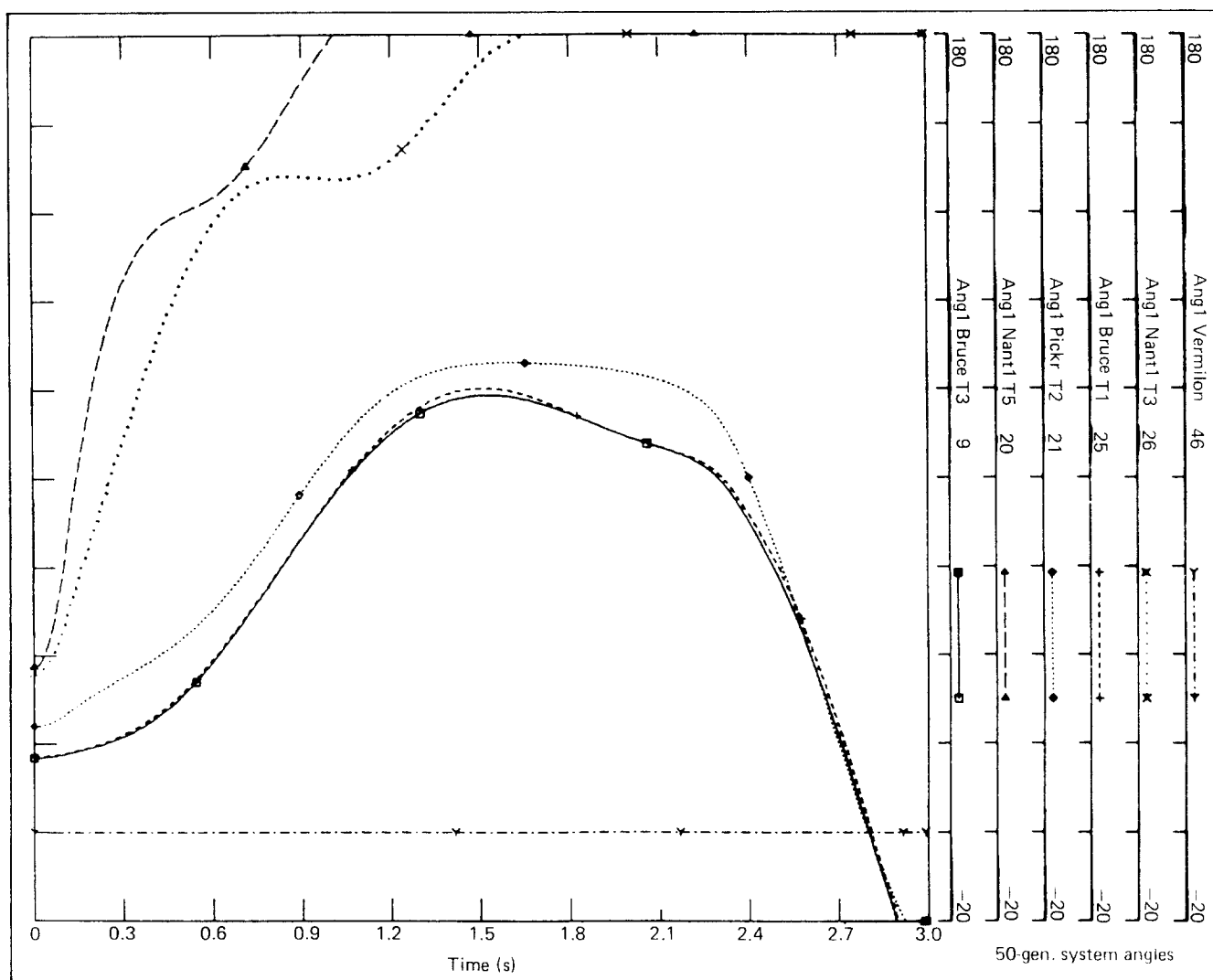


Figure 7. Machine angle plots of Nanticoke normal system (time-domain classical model; local MOI)

- calculates the generators' rotor angles and speeds at fault clearing;
- computes the generators' rotor angles for the post-fault network equilibrium conditions;
- determines the mode of disturbance and computes the generators' rotor angles at the controlling unstable equilibrium point (UEP);
- calculates the transient energy margin (unnormalized and normalized) and gives the transient stability classification.

VI.5.3 *TEF program input* The user supplies the following data:

- the load flow for the predisturbance conditions in IEEE common format;
- generator data (for the classical model only the inertia constant and transient reactance);
- switching data.

VI.5.4 *TEF program output*

- short and long outputs;

- identifies critical generators;
- gives the transient energy margin and its normalized value;
- gives components of the transient energy.

VII. Recent or current developments

Current development works in the TEF method, whether they are completed or in an advanced stage, aim at overcoming the current TEF method modelling limitations or at extending its application potential. Thus we begin this section by a discussion of the outstanding issues.

VII.1 Outstanding issues

VII.1.1 Modelling limitations

- higher-order generator models and modelling of the relevant generator controls, e.g. excitation system effects;
- non-linear load models – of particular interest would be the constant current load model;

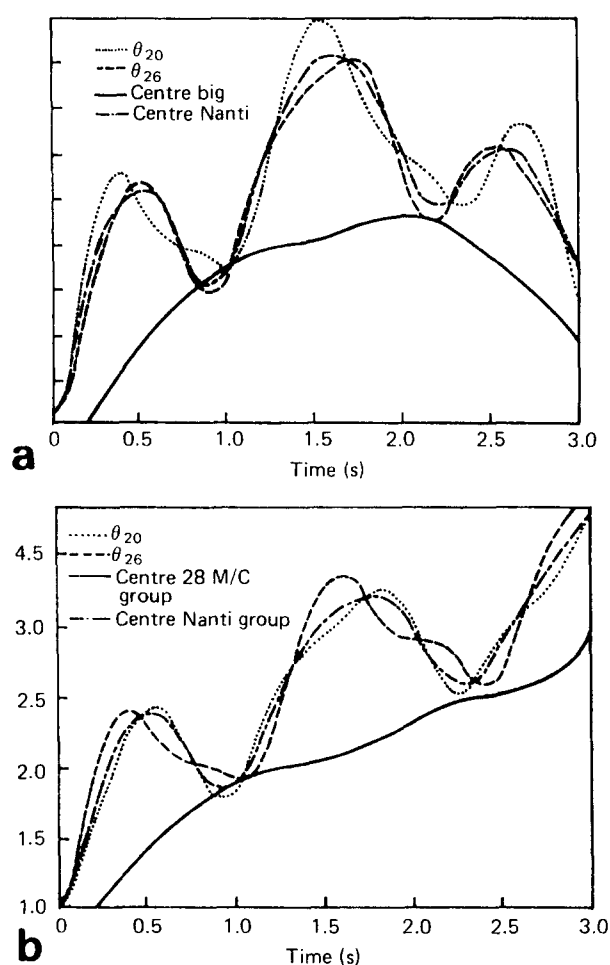


Figure 8. Nanticoke 3700 MW: (a) $t_{cl} = 0.12$ s; (b) $t_{cl} = 0.1221$ s

Table 1. Power limits comparison (from EPRI Report EL4980)

OH system	Power limit (MW)		
	Time simulation	TEF	Error
50-gen. Bruce	1481	1369	7.6%
100-gen. Bruce	1557	1514	2.7%
50-gen. Nanticoke (normal)	4027	3840.5	4.4%
50-gen. Nanticoke (inter-area mode)	3935	3800*	3.4%
SW Ontario, 115-gen. (inter-area mode)	4400	4450	1.1%

* Reference 7

- certain network elements, e.g. HVDC networks;
- certain network controls that are important in transient analysis, e.g. out-of-step relays, VAR controls, etc.

VII.1.2 First-swing limitation This is important if the TEF method is to be used in system planning applications. However, the authors feel that this is not as serious

a limitation as it originally was thought. Conditions leading to instability at later swings may be detected and identified in the first swing.

VII.1.3 Practical issues

- Reliability and accuracy are important in all applications; some additional testing of stressed system conditions may be needed.
- Speed; while considerable improvements have been achieved, faster solution is still needed for online applications.
- Size limitations; large networks may be needed to study certain dynamic problems; our experience is mostly with power system sizes of up to about 100 generators; more testing of larger-size networks is needed before we know the answer.
- To make the program more 'user-friendly' and to make the results more understandable and acceptable by the analyst; experience will be our guide.

VII.2 Incorporation of excitation system effects

VII.2.1 General approach The approach followed is based on the following assumptions⁸.

- When exciter effects are incorporated in a synchronous generator, the generator is represented by the two-axis model. The flux in such a generator is assumed not to change abruptly in the first swing; and if conditions (i.e. generator flux or internal EMF) initially and at the peak of the swing are known, only smooth variations between them take place.
- The first-swing transient is dominated by inertial effects. As a result of this assumption: (i) we are still concerned with the relative motion-of-inertia-dominant transients; and (ii) the potential energy boundary surface (PEBS) is considered elastic, and not stationary, for the change of E'_q and E'_d .
- The excitation system is represented by a one-time-constant, one-gain, one-limiter model. A special program is developed to obtain this equivalent model for most commonly available exciters, using the equivalent bandwidth technique for the frequency of interest.

VII.2.2 The new TEF⁸ Based on the approach outlined above, a value of the generator internal EMF E' is computed for a generator represented by the two-axis

Table 2. Typical solution times (with SNR)

Test system:	11-Gen. 55-Bus 183-Lin.	50-Gen. 145-Bus 649-Lin.	98-Gen. 1095-Bus 2041-Lin.
Reading data	0.55	1.22	4.37
Y-matrix	0.02	0.03	0.11
Reduction	0.39	1.21	2.94
SEP solution	0.04	0.54	2.22
UEP solution	0.03	0.53	2.23
Total (CPU s)	1.03	3.53	11.87

model. This value of E' is to be considered constant in the first swing, but its value gives the same system transient energy as for the actual situation with flux variation due to exciter effects. In the work done by the authors and their co-workers, an average value between the EMF at fault clearing and at the peak of the first swing is used.

An expression for the transient energy function has been developed for some of the generators represented by the *two-axis model with constant EMF* (estimated by the above procedure), and others represented classically.

VII.2.3 Procedure for transient stability assessment

Although the assumptions made in the approach outlined above considerably simplify the incorporation of exciter effects in the TEF method, a number of complexities remain and must be dealt with. Among them are: (i) whether the mode of disturbance (hence the critical generators) is affected by the changing EMF due to exciter effects; and (ii) estimation of conditions at the peak of the first swing is complicated by the fact that some of the derivatives may not be zero (i.e. in the limit the peak may not be at the UEP).

A procedure has been developed taking into account the above factors²¹. It has been tested on some medium-size test networks and is currently undergoing further testing.

Another issue which has been recently addressed is the number of generators which need to be modelled in detail (i.e. by the two-axis model) to account for the excitation system effects, since adding the detailed generator–exciter models considerably increases the computational burden. We have found that modelling only the critical generators in detail gives sufficiently accurate results²².

VII.3 Other model improvements

VII.3.1 Non-linear load models Two approaches are being pursued in parallel, apparently with success.

- Sparse formulation of the network equations. The primary motivation is to speed up network solutions at the SEP and the UEP²³ by avoiding network reduction and taking advantage of sparse techniques. It offers the added advantage of allowing non-linear load models. It is to be emphasized that in this approach the critical energy is still computed as the potential energy at the UEP.
- Non-linear loads, modelled by either constant current or constant power, are reflected at the internal generator nodes by appropriate sources. These are incorporated in the TEF expression. This work is at an advanced stage of development.

VII.3.2 Two-terminal HVDC A model for a two-terminal HVDC has been developed and incorporated in the TEF expression. The model is valid for first-swing transients and for faults in the AC network. Thus the inverter internal controls are neglected but the important external controls are included²⁴. This model is currently being incorporated in the TEF program software.

VII.3.3 Out-of-step relays See Section VI.4.

VII.4 Emerging applications

A brief review of some of the new applications of the TEF method which are either under investigation or in the development stage is given below.

VII.4.1 Use as a screening tool

- For planning studies of multi-swing transients.
- To identify, in a stream of runs containing many (typically hundreds) cases, the critical cases to study in detail by conventional stability programs.
- Stressed conditions likely to cause system separation at a later swing can be identified in the first swing by TEF analysis.
- Symptoms of the stressed conditions are: low energy margin (caused by high loading levels); low value of critical energy (indicative of weak transmission); presence of inter-area mode.

VII.4.2 Sensitivity coefficients The sensitivity of the transient energy margin to changes in power at certain plants or transmission interfaces is under active investigation. Analytical expressions are being sought which have greater range of validity than those previously developed with linearized techniques. The application potentials are:

- stability-limited plant loadings;
- stability-limited transmission flow limit;
- development of safe regimes of operation when several key parameters are involved;
- development of strategies for corrective action when contingencies occur;
- determination of which contingencies to check (for second-contingency evaluation).

VII.4.3 Online detection of instability For simple system configurations, where the impedance 'seen' by a critical generator does not substantially change (with normal switching procedures), it is possible to make an (approximate) estimate of V_{cr} . When the transient energy obtained by actual measurement of generator angles and speeds approaches V_{cr} , detection of instability is possible.

VII.4.4 Online derivation of operating security limits This scheme is now under development for installation at the Ontario Hydro Energy Management System. Its objective is to continuously manage the stability of (part of) the Ontario Hydro power network within defined security criteria. A combination of offline studies and online analysis of some (20–30) contingencies will be used to determine safe regimes of operation. It will replace a recently developed scheme based on the equal-area criterion^{25,26}.

The online scheme is designed to be able to determine in real time, and under changing system dispatch and planned or forced equipment outages, whether the stability criteria are met. The steps involved are: contingency and post-disturbance selection; computation of stability limits; identification and display of security regimes; detection and alarming out-of-security regime operation; and taking corrective action.

Development work in this area is under way. It is planned for completion in 1990.

VII.4.5 Further applications In addition to the above, the authors believe that the following applications of the TEF method are possible with today's technology:

- reliability evaluation of stability-limited networks, in which the probabilistic variation of certain parameters and the energy margin sensitivity to these parameters are to be used;
- use of expert system/artificial intelligence techniques in or near real time to identify important contingencies to be investigated for assessment of system security.

VIII. Conclusions

Examining the list of contributions to the TEF literature in the last few years gives the reader an indication of the rate at which the area of direct stability analysis is progressing.

There is no question in the mind of the authors that the TEF method, whether in its present form or in a modified version, is here to stay as one of the important tools of stability analysis available to industry engineers. Yet, a host of other questions immediately present themselves. Some can be answered with a reasonable degree of certainty; others require a good measure of guessing; and some cannot be answered at all. Instead of the authors addressing each question, some general observations will help each of us find the answers.

The interconnected power network of North America has been increasingly concerned with power system stability in recent years. Because of heavier transmission loadings, larger generator unit size, increased dependence on controls, lesser use of oil- and gas-fired plants, etc, stability-limited conditions do exist in many parts of the interconnection, either as a first or second contingency.

This trend does not appear to be on the decline. On the contrary, in addition to all the above factors, the newly added effect of 'deregulation' and 'competition' will tend to accentuate the conditions causing the transmission network to be stability-limited.

Meeting NERC criteria for security and for preventing cascading outages in stability-limited conditions is no longer of concern *only* to the designer. Rather, for several power networks it has become an operating concern. This puts new constraints on transient stability analysis: in the number, frequency and types of stability studies to be made; in the time period available to run them; and in the type of answers required.

The industry has now at its disposal an excellent tool for transient analysis, i.e. the transient stability computer program. To meet some of the above constraints, this tool should be augmented with other tools. In other situations, e.g. in or near real time applications, the conventional tool needs to be replaced. As shown in the previous sections, the TEF method is a good candidate for these applications. This method's development has reached a degree of maturity which deserves acceptance by the industry. Only then can the potential of the TEF method be realized.

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